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6 March 2026

Dr. Patrick Maschmeyer
AG Ludwig – Stem Cell Dynamics and Mitochondrial Genomics
Max Delbrück Center for Molecular Medicine (MDC)
Hannoversche Straße 28
10115 Berlin

Application: Bioinformatician / Computational Scientist – Ludwig Lab

Dear Dr. Maschmeyer,

I am writing to express my strong interest in joining the Ludwig Lab as a Bioinformatician / Computational Scientist. My course coordinator at CQ Beratung + Bildung, Nassim, recently shared that you are looking for candidates who can develop computational tools and apply AI/ML methods. Given the lab's focus on single-cell multi-omics and the computational demands of maintaining toolkits like `mgatk` or pipelines like `PUMATAC`, my transition from a Ph.D. in Astrochemistry to bioinformatics pipeline engineering makes me a highly suitable candidate.

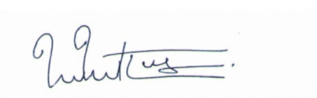
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics. At my current internship with HealthTwiSt GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility (`identical()`), and reduced the 1,834-line codebase by 26%, dramatically improving maintainability.

The Ludwig Lab's recent work—particularly the large-scale statistical analysis of the UK Biobank dataset and high-throughput methods like PERFF-seq—requires exactly this kind of rigorous data engineering and statistical modeling. I bring a strong foundation in the tools your lab relies on:

- **Pipeline Development:** Highly proficient in R (Tidymodels, Bioconductor) and Python (Pandas, Scikit-learn). I build reproducible workflows and maintain extensive codebases.
- **Large-Scale Data & Statistics:** Processed terabytes of raw experimental data during my Ph.D. and developed C++/Fortran algorithms. My expertise in multivariate statistics and bootstrap subsampling strongly aligns with extracting signal from high-dimensional datasets.
- **AI / Machine Learning:** Actively expanding my ML capabilities to integrate multi-omic layers and infer cell trajectories. Alongside building ML workflows at HealthTwiSt, I am completing IBM's "Machine Learning with Python" and "Introduction to AI" certifications (Scikit-learn, clustering).

I am deeply motivated by the Ludwig Lab's mission to decode mitochondrial genetics. My strong quantitative foundation and recent pipeline engineering experience will allow me to immediately contribute to your computational infrastructure. I welcome the opportunity to discuss further and am available for an interview.

Hamburg, 6 March 2026



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